

The witness party

Unique completion as a quantization condition

Draft. Not a submission.

There is a small formal question inside an old philosophical one. The old question is whether a later event can be necessary for an earlier one. The small question is whether the classical maps that already refuse nonsense can be read as quantum processes. The small answer is yes for finite alphabets, once a source and a sink are allowed, if a contraction is granted. That does not answer the old question.

1. Why the word is not the problem

Dummett set the dictionary objection aside in 1954 and 1964. Black: an agent who has seen the earlier event can try to prevent the later one. Lewis splits the modal and does not select the histories. Horwich asks what happens to the intervention. Faye, Stanford Encyclopedia, revision 28 October 2025, still treats the consistency paradoxes as unresolved. Only consistent loops occur can mean a discard, a filter, or a definition of the admitted set. The third reading can be failed. The identity loop fails it. The swap fails it.

2. The billiard

Godel 1949 has closed timelike curves. Morris, Thorne and Yurtsever convert a wormhole into a time machine if the throat is held by exotic stress-energy. Echeverria, Klinkhammer and Thorne find one continuation, two, or none. A process function asks for one continuation for every intervention. The billiard has the wrong multiplicity. Hawking's chronology protection is a different question.

3. Equations

(1) $o = f(w(o))$. The output returns itself.

(2) $w(a, b, c) = (\text{not } b \text{ and } c, \text{not } c \text{ and } a, \text{not } a \text{ and } b)$. Two-to-one.

(3) (o, e) maps to $(w(o) \text{ XOR } e, o)$.

(4) (i, s) maps to $(s, i \text{ XOR } w(s))$.

On bits exclusive or cancels. Lean checked both compositions.

(5) some party disagrees in the output and agrees in w . Checked.

(6) $\delta = w(s') \text{ XOR } w(s)$.

(7) $M \text{ M-dagger} = I$. Written, not kernel-checked.

4. Lean

Checked in lean/Full.lean: the disagreement lemma and the bit bijection. Not checked: the gram identity. The upload of the eleven-page file with embedded equation images failed because the repository connection accepts file content as text and truncates or corrupts bytes above 127. This file is ASCII so the download can match.

5. Embeddability

A channel embeds as free propagation. Lugano violates a causal inequality. Post-selection is a simulation. The bijection is reversibility. No metric for Lugano is claimed.

References

Dummett 1954, 1964. Black 1956. Lewis 1976. Horwich 1987. Faye, SEP, 28 October 2025. Godel 1949. Morris, Thorne, Yurtsever 1988. Echeverria, Klinkhammer, Thorne 1991. Earman 1995. Deutsch 1991. Baumeler, Costa, Ralph, Wolf, Zych 2019. Araujo et al. 2017. Leifer and Pusey 2017. Dourdent et al. arXiv:2609.39735. Stinespring 1955.